

Fig. 5.12

5.4 Controller

Controller is used to control the torque and speed of the electrical drives.

a. AC - DC controller

AC / DC controllers are of electronic system. A steady state speed and voltage of the device can be used to convert it to the desired limit. This allows to control the speed, rotation, horse power, and direction of rotation. It is used more effectively because of the energy savings of the drive.

b. DC - DC controller

The control of the DC - DC driver is used for heavy load traction. The speed control and rotation direction is essentially modified based on the voltage level normally connected to a fixed power voltage supply.

5.5 Electric vehicles

5.5.1 Two wheeler electric vehicle

The UK-based Humming bird company discoverd weight less a two-wheeler as shown in figure 5.13 on the lowest electric bike in the world. This bike will ride at a speed of 25 km, and can drive up to 30 km for single charge. This vehicle, like a bicycle is also used as a smart bike. The weight of the smart bike is 10.3 kg.



Fig. 5.13 **Two wheeler electric vehicle**

5.5.2 Electric vehicle



Fig. 5.14 **Electric vehicle**

Increasing air pollution in India, there are a number of attempts were going on to run vehicle with electricity is as source instead of petrol or diesel. The hyper car type in figure 5.14 is enhanced by the power of the turbine power that does not affect the environment. Such a system is a rare use in electric vehicles. The top of the cover is made of very light metal. The hyper car has been developed with all the features of beautiful design and smooth running.

5.5.3 Solar vehicle

As shown in figure 5.15, the solar vehicle is a vehicle that runs with solar

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